

AP EAMCET MEDICAL

Shift: 2

Year: 2018

No. of Question: 160

BOTANY

1. His characteristic feature of Ectocarpus is

- I. Green filamentous algae with simple branches
- II. Cell contain chlorophyll a, d. carotenoids and xanthophylls
- III. Life cycle is haplodiplontic
- IV. Zoospores are pyriform with two flagella attached laterally

Options

- a) I, II
- b) I, III
- c) III, IV
- d) II, IV

2. Choose the correct group of organisms based on their cell wall materials respectively Cellulose, Peptidoglycan, Algin, Mycolic acid and Pseudomurein.

Options

- I. Claviceps, Bacillus, Ulothrix, Corynebacterium, Laminaria
- II. Ulothrix, Bacillus, Laminaria. Corynebacterium, Archaeobacteria

III. Laminaria. Archaeobacteria, Ulothrix. Bacillus.
Corynebacterium Bacillus. Corynebacterium, Ulothrix.
Laminaria, Archaeobacteria

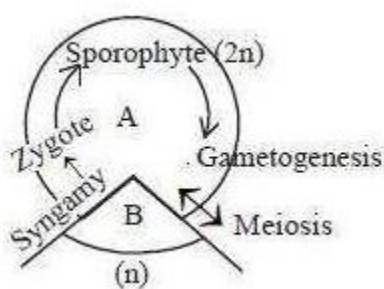
3. Study the following statements

- i. Branched chain of proteins are present in membranes of Archeobacteria
- ii. Histones are absent in Noctiluca chromosomes
- iii. Asexual spores of diatoms are Auxospores
- iv. In Nostoc Chromatin is present in centrioplasm

Choose the correct answer
options

- a) i,ii,iii,iv
- b) i,iv
- c) i,iii
- d) ii,iv

4. Identify correct answer with reference to life cycle of Algae given above



Options

- a) It is haplo-diplontic life cycle of Laminaria
- b) Sporophyte is dependent on Gametophyte
- c) Sporophyte and Gametophyte are independent
- d) It represents life cycle of focus

5. Fruits with Papery mesocarp, Cartilagenous endocarp and undifferentiated endocarp are respectively

Options

- a) Citrus, Apple, Guava
- b) Apple, Cucumber, Grapes
- c) Apple, Cucumber, Tomato
- d) Cucumber, Citrus, Apple

6. Centripetal, Aacropetal and irregular arrangement of sessile flowers are seen in the inflerescences of these plants respectively

Options

- a) Tridax. Musa and Ficus
- b) Achyranthus, Cocos and Leucas
- c) Carrot, Colacasia and Euphorbia
- d) Hamelia, Crotalaria and Solanum

7. The steps to be practiced for artificial hybridization in two female parents having unisexual and bisexual flowers

Options

Unisexual flowers	Bisexual flowers
a) Bagging. Debagging. Emasculation	Emasculation. Bagging. Pollination
b) Bagging, Emasculation. Pollination	Bagging. Rebagging. Pollination
c) Emasculation. Rebagging. Pollination	Emasculation. Rebagging. Bagging
d) Bagging. Pollination. Rebagging	Emasculation. Bagging. Pollination. Rebagging

8. Find the incorrect statements

- A. Lower plants shift to sexual method of reproduction just before the onset of a adverse conditions takes place
- B. In South India, hilly areas transform into blue stretches by the flowering of Agave and Bamboo
- C. Single cell Zygote is the vital link that ensures continuity of the species from one generation to the next
- D. In Angio sperm pollen grains are shed a 3-celled stage

Options

- a) A,B
- b) C,D
- c) A,C
- d) B,D

9. Identify the correct sequence of events that occur during development of Embryo sac

i. Reduction division	ii. Embryosac
iii. Nucellus	iv. Linear tetrad
v. Megaspore mother cell	vi. Functional megaspore
vii. Mitotic divisions	

Options

- a) e,c,a,d,f,g,b
- b) c,e,a,d,f,g,b
- c) c,e,g,d,f,a,d
- d) c,e,a,f,g,d,b

10. Mention the ploidy of parthenogenetic embryo, parthenocarpic fruit and apomictic seeds in series options

- a) $n, 2n, 2n$
- b) $2n, n, 2n$
- c) $n, 2n, n$
- d) $2n, n, n$

11. Find the correct floral formula from the given floral description
 Bracteate. ebracteolate. pentamerous. bisexual, actinomorphic, sepals 5, polysepalous, imbricate aestivation, petals 5, gamopetalous, twisted, aestivation, stamens 5, epipetalous: pentacarpellary syncarpus. superior ovary with axile placentation

options

- a) $\text{Br, Ebrl, } \oplus, \text{ } \overline{\text{K}}_5, \text{ } \overline{\text{C}}_5, \text{ } \overline{\text{A}}_5, \text{ } \overline{\text{G}}_{(5)}$
- b) $\text{Br, brl, } \oplus, \text{ } \overline{\text{K}}_5, \text{ } \overline{\text{C}}_5, \text{ } \overline{\text{A}}_5, \text{ } \overline{\text{G}}_{(5)}$
- c) $\text{Br, brl, } \oplus, \text{ } \overline{\text{K}}_{(5)}, \text{ } \overline{\text{C}}_5, \text{ } \overline{\text{A}}_5, \text{ } \overline{\text{G}}_{(5)}$
- d) $\text{Br, Ebrl, } \oplus, \text{ } \overline{\text{K}}_5, \text{ } \overline{\text{C}}_{(5)}, \text{ } \overline{\text{A}}_5, \text{ } \overline{\text{G}}_{(5)}$

12. Match the following

List-I	List-II	List-III
a) Interphase	i) Synapsed homologous Chromosomes	I) Two sequential cycles

b) Meiosis	ii) Crossing over	II) Exchange of genetic material
c) Bivalent	iii) Single cycle of DNA Replication	III) 95% of cell cycle duration
d) Recombinase	iv) Between two M-phases	IV) Zygotene

Options

	A	B	C	D
a)	iv,III	iii,I	i.IV	ii,II
b)	iv,III	i,I	ii.II	iii,IV
c)	iii,III	I,II	ii,I	iv,IV
d)	iv,III	i,II	ii,IV	iii,I

13. Choose the correct statement

- A. Mesosomes are exterior extensions of plasma membrane outside the cell
- B. In prokaryotic cells filamentous extensions of cell walls are called flagella
- C. The membrane of erythrocyte has nearly 70 % protein and 20% lipids
- D. The Mitochondria. Chloroplast and peroxisomes are not co-ordinated with endomembrane system

Options

- a) B,D
- b) C,D

- c) B,C
d) A,D

14. Choose the correct statements

- I. Inulin is a polymer of glucose
II. Cellulose does not contain complex helices
III. Starch is a polymer with helical secondary structure
IV. Store house of energy in animal tissue is glycogen

Options

- a) I,IV,III
b) I,III,II
c) II,IV,I
d) II,III,IV

15. Arrange the following four levels of protein structures
– Primary, Secondary, Tertiary and quaternary in a correct match

I. Primary structure	First Terminal - C Last terminal - N	1 Polypeptide
II. Secondary structure	Right handed Helic polypeptide	1 Polypeptide
II. Tertiary structure	Long chain folded like a ball	2 Polypeptides

V. Quaternary structure	Protein subunits present	More than one Polypeptides
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Choose the correct combination

Options

- a) I,III
- b) II,IV
- c) III,IV
- d) I,IV

16. Functions not associated with peroxysomes

- i. Conversion of stored lipids to carbohydrates
- ii. Catabolism of fatty acids
- iii. Involve in photophosphorylation
- iv. Production of Hydrogen peroxide
- v. Synthesis of phospholipids

The correct answer is

- a) i,iii,iv
- b) ii,iii,iv
- c) ii,iv,v
- d) iii,v

17. Choose the functions of cytoskeleton

- i. Formation of cell plate during cell division
- ii. Mechanical support
- iii. Cell motility
- iv. Intercellular transport

The correct answer is

Options

- a) I,III
- b) II,IV
- c) III,IV
- d) II,III

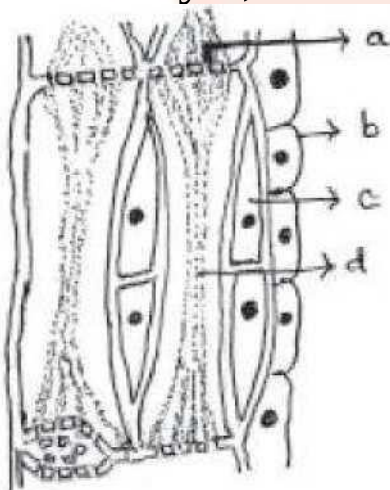
18. Assertion (A): Grasses get regenerated even after grazed by herbivores

Reason(R): In grasses meristems are present between mature tissues

Options

- a) (A) and (R) are true and (R) is the correct explanation of (A)
- b) (A) and (R) are true and (R) is not the correct explanation of (A)
- c) (A) is true, but (R) is false.
- d) (A) is false, but (R) is true.

19. Identify c, d



Options

- a) Sieve pore. Phloem Parenchyma
- b) Phloem Parenchyma. Companion cell
- c) Companion cell. Sieve tube

d) Sieve pore, Sieve tube

20. Assertion (A): The stability of coupled system formed by climate and natural ecosystems is an important ecosystem services.

Reason (R): Natural ecosystems prevent overstating of the earth by removing more of the greenhouse gases from atmosphere

Options

- a) (A) and (R) are true and (R) is the correct explanation of (A)
- b) (A) and (R) are true but (R) is not the correct explanation of (A)
- c) (A) is true; but (R) is false
- d) (A) is false, but (R) is true

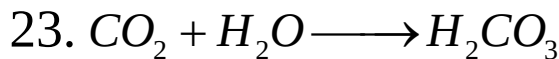
21. Find the correct answer

- a) Water movement is against concentration gradient in Imbibition
- b) Addition of solute particles to a solution enhances water potential
- c) Rate of osmosis depends on both the pressure gradient and concentration gradient
- d) Pressure potential is negative in sieve tubes

22. Select the correct answer

- a) Absorbed Nitrogen is accumulated by the plants in the form of Ammonium ions
- b) In free living conditions nitrogenase is not operational

- c) Nodules of soyabean export the fixed nitrogen as amides
- d) Amino group of Aspartic acid is replaced by hydroxyl group to form asparagine



Number of H_2CO_3 molecules released in the above reaction in the Absence and presence of Carbonic anhydrase enzyme respectively

Options

- a) 2000 / hr and 6,00,000/hr
- b) 2000/hr and 60,000/hr
- c) 200 /s and 60,000/s
- d) 200/hr and 6,00,000/s

24. Functions of lamellae of grana and stroma lamellae of chloroplast during photo phosphorylation

Options

- a) Both release $NADPH_2$ and ATP
- b) Former releases $NADPH_2$ and ATP, later splits water
- c) Former splits water, later produce ATP
- d) Both splits water and release ATP

25. The ratio of NADP, H_2O and electrons required to evolve $4O_2$ molecules through photophosphorylation are

Options

- a) 1:1:2
- b) 1:2:4
- c) 1:1:2

d) 1:2:8

26. Number of compounds with C_2 and C_4 intermediate substances appear in TCA cycle, when one glucose molecule oxidized through respiration

Options

- a) 1,8
- b) 2,10
- c) 1,6
- d) 2,8

27. In 138 hours of time. two leaves A and B with 5 cm^2 and 50 cm^2 of size show absolute increase to 10 cm^2 and 55 cm^2 respectively. Then the leaf which shows highest relative growth rate is

Options

- a) Only A
- b) Only B
- c) Both A & B do not show any difference in relative growth
- d) Both A & B do not show any difference in absolute growth

28. Functions not induced by Cytokinins

- i. Production of leaves
- ii. Production of chloroplasts
- iii. Nutrient mobilisation
- iv. Leaf senescence
- v. Xylem differentiation
- vi. Synthesis of Carotenoids

Options

- a) i, ii, iii
- b) iv, v, vi
- c) ii, iv, v
- d) iii, iv, vi

29. Identify incorrect statements

- i. Chlorobium cannot utilize radiant energy
- ii. Nitrosomonas derives energy from sunlight
- iii. Rhodospirillum derives carbon from carbondioxide
- iv. Bdellovibrio bacteriophorus grows as a parasite on some harmful viruses

Options

- a) ii,iii
- b) i,ii,iii
- c) i,ii,iii,iv
- d) i,ii

30. HIV belongs to the Genus

Options

- a) Retrovirus
- b) Lentivirus
- c) Poliovirus
- d) Rotavirus

31. Identify incorrect statement with reference to hybridisation experiment in lentil

Options

- a) Genotypic ratio of F_2 generation is identical to that of the ratio of Mendel monohybrid cross

- b) Law of dominance is not applicable to F₁ generation
- c) Neither of the alleles are expressed in F₁ hybrids
- d) Both alleles are expressed independently in F, hybrids

32. The character of RNA is

Options

- a) Mutate at faster rate
- b) Non catalytic
- c) Thymine is present at the place of uracil
- d) RNA is chemically less reactive and structurally more stable when compared to DNA

33. Role of E.coli in replication of DNA is

- i. Semi conservative replication was proved first
- ii. Polymerises approximately 2000 bp per second
- iii. Replication time is 38 minutes
- iv. Length of DNA is 6.6×10^9 bp

The correct answer is

Options

- a) i,ii,iii,iv
- b) ii,iii,iv
- c) ii,iv
- d) i,ii,iii

34. Identify the incorrect statement

Options

- a) Factory of protein synthesis is ribosome
- b) Cell contain 64 types of tRNAs
- c) Because, of nucleotide 2'-OH group RNA is easily degradable

- d) Deoxyribonucleoside triphosphates provide energy for DNA polymerisation

35. Key tools of r-DNA technology are

- I. Molecular scissors
- II. P^{BR} 322
- III. Gene gun
- IV. TV Bioreactor

Options

- a) I,III
- b) I,IV
- c) I,II
- d) II,III

36. Genetically transformed host cells can be selected by

- i. Antibiotic resistant genes
- ii. Radioactive ds DNA having complementarity to the whole length of desired DNA
- iii. Coding sequence of β – galactosidase
- iv. Radioactive ss RNA having complementarity to at least one part of desired DNA
- v. Radioactive ss RNA having complementarity to the whole length of desired DNA

Options

- a) i,ii,iv
- b) ii,iii,iv
- c) i,ii,iii,v
- d) i,iii,iv

37. RNA interference (RNAi) is a mechanism That
Options

- a) Prevents translation of specific mRNA
- b) prevents transcription of mRNA
- c) Prevents replication of DNA
- d) Si RNA produced by ds RNA, cuts DNA into pieces

38. Excellent biocontrol agents to treat ecologically sensitive areas

Options

- a) Nucleopolyhedrovirus
- b) Trichoderma
- c) Bacillus thuringiensis
- d) Azospirillum

39. Microbes not used in house hold products

- I. Trichoderma polysporum
- II. Rhizobium
- III. Closteridium butylicum
- IV. Propionibacterium shrmami

Options

- a) I,IV
- b) I,II,III
- c) II,IV
- d) I,IV

40. Biofortified vegetable crops released by IARI

- I. Vitamin 'A' enriched bathua and Tomato
- II. Vitamin 'C' enriched Carrot and Spinach
- III. Iron and Calcium enriched Spinach and bathua

IV. Protein enriched French bean and garden peas

Options

- a) III,IV
- b) I,III
- c) II,IV
- d) II,III

ZOOLOGY

41. Assertion (A): Specific endangered species are protected in sanctuaries

Reason (R.): Protection of animals in their natural habitats is called in-situ conservation

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

42. Study the following statements regarding tissues

- A. Basophils constitute least percentage of total granulocytes
- B. Muscles attached to hair-follicle in skin are striated muscles
- C. Os cordis is an endochondral bone

D. Mucous connective tissue is present in umbilical cord

Options

- a) A&C
- b) C&D
- c) B&D
- d) B&D

43. Assertion (A): Echinoderms show pentamerous radial symmetry

Reason (R): Animals with radial symmetry respond equally to stimuli that arrive from all directions

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

44. Study the following statements

- A. Cloacal respiration and two jawed pedicellariae are characteristic to sea cucumbers
- B. Epidermal cells of Digesta have rhabdites
- C. Colloblasts help in food capture in Pleurobrachia
- D. Maxillae are modified into gnathochilarium in Scolopendra

Options

- a) A&B
- b) B&C
- c) B&D
- d) A&C

45. Match the following

Animal	Character
A) Neomania	I) Single shell plate
B) Dentalium	II) Two pairs of ctenidia
C) Neopilina	III) Cuticle with spicules
D) Nautilus	IV) Captacula
	V) Crystalline style

Options

- | | | | | |
|----|-----|-----|-----|----|
| | A | B | C | D |
| a) | I | II | III | V |
| b) | III | II | IV | I |
| c) | I | III | IV | V |
| d) | III | IV | I | II |

46. Study the following and choose the correct combinations

S.No.	Group	Character	Example
I.	Tunicata	Bioluminescence	<i>Pyrosoma</i>
II.	Osteichthys	Pelvic fins with claspers	<i>Echeneis</i>
III.	Mammalia	Six cervical vertebrae	<i>Bradypus</i>
IV.	Caecilians	Amphicoelous vertebrae	<i>Ichthyophis</i>

Options

- a) I,II
- b) II,III
- c) III,IV
- d) I,IV

47. Which of the following characters are found in ratitae

- I. Males with penis
- II. Sternum with keel
- III. Absence of clavicle
- IV. Presence of syrinx
- V. Discontinuous distribution

Options

- a) I,II,IV
- b) I,III,V
- c) II,III,V
- d) III,IV,V

48. Match the following

List -1	List - II
A. Giardia	I. Many
B. Trypanosoma	II. 3 pairs
C. Trichomonas	III. 4 pairs
D. Trichonympha	IV. Four
	V. Single

The correct answer is

Options

A B C D

- a) III IV V I
- b) III V II IV
- c) III V IV I
- d) III II IV I

49. Study the following statements regarding drugs

- A. Benzodiazepines are used as painkillers
- B. Smack slows down the body functions
- C. The opioid drug obtained from poppy plant affects cardiovascular system
- D. Crack interferes with the synthesis of dopamine

Choose the incorrect statements from the above

Options

- a) A, D
- b) B, C
- c) A, C
- d) B, D

50. Match the following

List-I	List-II
A. Neoplasia	I. Increase in number of cells
B. Hyperplasia	II. Degeneration of gonads of host
C. Hypertrophy	III. Increase in the size of the host
D. Parasitic castration	IV. Formation of a new structure in host tissue
	V. Abnormal increase in volume of host cell

The correct answer is

Options

- | | A | B | C | D |
|----|-----|----|----|----|
| a) | IV | V | I | II |
| b) | III | IV | V | II |
| c) | IV | I | V | I |
| d) | III | I | IV | II |

51. Assertion (A): Ring worm is a fungal disease

Reason (R): The mode of infection is by inoculation

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

52. The following are types of parasites

- I. Intracellular
- II. Coelozoic
- III. Intercellular
- IV. Hyperparasite

Which of the above are correct regarding *Nosema notabilis*

Options

- a) I,II

- b) I, IV
- c) II, III
- d) III, IV

53. The sense organs present on proximal parts of tibia in Cockroach are sensitive to

Options

- a) Sound vibrations
- b) Taste
- c) Movement of flagellae of antennae
- d) Ground vibrations

54. The following are the ganglia of autonomous nervous system of Cockroach

- I. Proventricular ganglion
- II. Frontal ganglion
- III. Ingluvial ganglion
- IV. Occipital ganglion

Arrange them in sequence from anterior part

Options

- a) II, I, III, IV
- b) IV, II, III, I
- c) II, III, IV, I
- d) II, IV, III, I

55. Assertion (A) : Algal blooms are formed in fresh water bodies

Reason (R): Eutrophication

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)

- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

56. Study the following statements

- I. Montreal protocol is aimed at the safe disposal of radio active wastes
 - II. The association of honey bee and orchid flower is commensalism
 - III. At high altitude the red blood cell production increases in man to compensate low oxygen availability
 - IV. The pyramid of biomass in sea is inverted
- Correct statements are among the above

Options

- a) I,III
- b) III,IV
- c) II,III
- d) I,IV

57. Match the following

Animal	Adaptation
A. Seals	I. Salt elimination by nasal secretions
B. Sea gulls	II. Metabolic water
C. Kangaroo Rat	III. Blubber
D. Turtles	IV. Glomerular Kidney

	V. Salt elimination by tears
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The correct answer is

Options

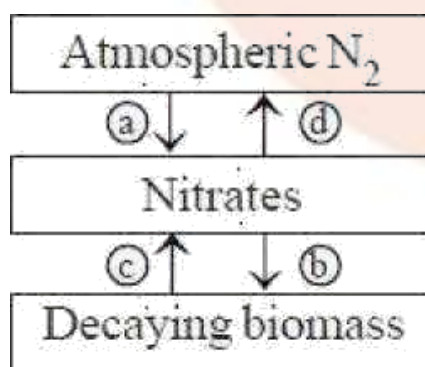
- | | A | B | C | D |
|----|-----|---|----|----|
| a) | II | V | IV | I |
| b) | III | I | II | V |
| c) | II | I | IV | V |
| d) | III | I | IV | II |

58. Which of the following factors faster decomposition of detritus

Options

- a) Lignin and moist environment
- b) Nitrogen and anaerobic environment
- c) Nitrogen and sugars
- d) Chitin and Lignin

59. Study the following model of Nitrogen cycle



Identify the concerned bacterium in step (D)

Options

- a) Nitrosomonas
- b) Nitrobacter

- c) Nitrosococcus
- d) Thiobacillus

60. Study the following and choose parts related to larynx

- A. Rimaglottidis
- B. Turbinals
- C. Cartilages of Santrini
- D. Adam's apple
- E. Soft palate

Options

- a) A,C,D
- b) B,D,E
- c) A,C,E
- d) A,B,D

61. Study the following respiratory volumes (*ml*) [Based on maximum values]

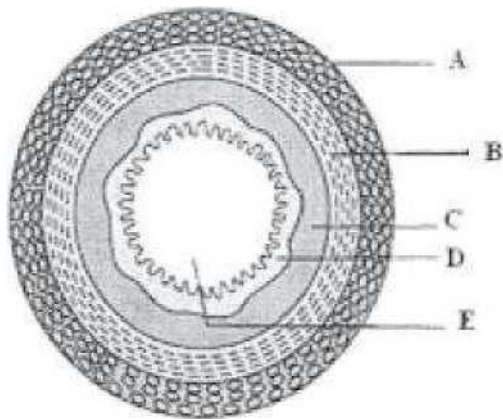
- I. Inspiratory reserve volume
- II. Tidal volume
- III. Inspiratory capacity
- IV. Residual volume
- V. Expiratory reserve volume

Arrange them in collect sequence in decreasing order

Options

- a) IV,III,II,I,V
- b) II,I,III,IV,V
- c) I,III,IV,V,II
- d) III,I,IV,V,II

62. Identify the part labelled as D in the above diagram of T.S. of human alimentary canal.



Options

- a) Mucosa
- b) Serosa
- c) Sub mucosa
- d) Muscularis mucosa

63. Study the following statements regarding circulation

- A. Atherosclerosis is a congestive heart failure
- B. The walls of lymph vessels contain smooth muscles
- C. Ligamentum arteriosum connects systemic and carotid arches
- D. Activation of proconvertin results in the formation of prothrombin activator

Choose the correct statements from the above

Options

- a) B,C
- b) A,D
- c) A,C
- d) B,D

64. Assertion (A): Uric acid is produced in mammals by purine metabolism

Reason (R): Uricase converts uric acid into allantoin in human beings

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

65. Choose the incorrect combinations regarding urine formation

	Part of Nephron	Selective reabsorption	Secretion
A	Distal convoluted tubule	Facultative reabsorption of Na^+ and H_2O	H^+ , K^+ and NH_4^+
B	Descending limb of loop of Henle	H_2O by passive transport	Impermeable to electrolytes
C	Proximal convoluted tubule	H^+ and K^+ ions	H^+ and urea
D	Collecting duct	K^+ and H_2O	Ammonia and HCO_3^-

Options

- a) A&C

- b) C&D
- c) B&C
- d) A&D

66. which of the following is incorrect combination?

Options

- a) scapula-Triangular
- b) sphenoid bone-keystone
- c) Hyoid bone-‘U’ shaped
- d) Sacrum-Cup shaped bone

67. Identify the part labelled as B in the above diagram of thin filament of a myofibril

Options

- a) Tropomyosin
- b) Troponin
- c) Actin
- d) Myosin

68. Study the following statements

- I. Somatotropin and somatostatin are released from hypothalamus and are antagonistic to each other.
- II. Cushing's syndrome is characterised by glucosuria and formation of ketone bodies.
- III. Parathyroid hormone is a protein.
- IV. Catecholamines and cortisol stimulate gluconeogenesis.

Correct statements are

Options

- a) I,III
- b) II,IV

- c) I,II
d) III,IV

69. Match the following

List -1	List-II
A. Anti-Tetanus serum	I. Artificial active immunity
B. Cytokines	II. Natural passive immunity
C. Colostrum	III. Artificial passive immunity
D. Tetanus Toxoid	IV. Natural active immunity
	V. Innate immunity

The correct answer is

Options

- | | | | |
|--------|----|----|-----|
| A | B | C | D |
| a) III | IV | V | II |
| b) III | II | IV | I |
| c) I | IV | II | III |
| d) III | V | II | I |

70. The events observed by the end of second month of pregnancy are

Options

- a) Separation of eye lids
b) Development of limbs and digits
c) Appearance of hair on head

d) Formation of eye lashes

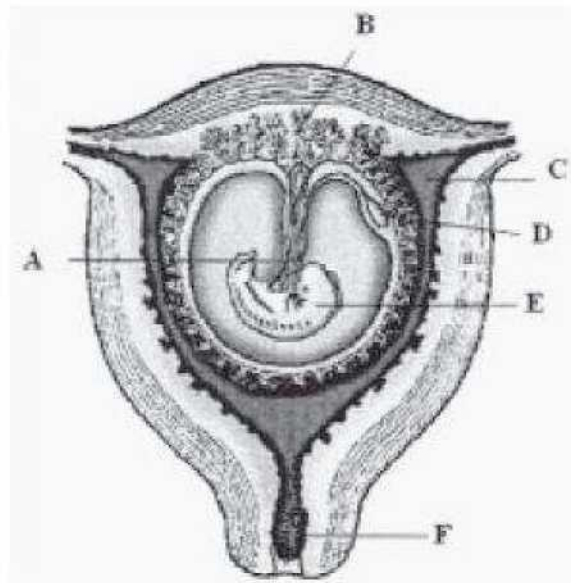
71. Choose the incorrect combinations from the following

S.No.	List -1	List - II	List - III
I.	Skene's glands	Homologous to Cowper's glands	Lubricates vagina
II.	Colostrum	Presence of IgA antibodies	Protection of new born from initial infections
III.	Hyaluronidase	Released by aerosome	Dissolves zona pellucida
IV.	Hypomere	Somatic splanchnic and mesoderms	Gives rise to pericardial cavity

Options

- a) II,IV
- b) I,III
- c) I,IV
- d) II,III

72. Identify the pan marked as D in the above diagram



Options

- a) Amnion
- b) Allantois
- c) Chorion
- d) Yolk sac

73. Assertion (A) : Male ants develop by parthenogenesis

Reason (R) : Male ants produce sperms by mitosis

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

74. study the following statements

- A. The alleles for Rh antigen are located on different location chromosome 9

- B. Three alleles of a single gene express ABO blood types in man
- C. The erythroblastosis foetalis is caused due to IgM type antibodies
- D. Isoagglutinogens are glycolipids

Correct the incorrect statements from the above

Options

- a) B&D
- b) A&B
- c) A&C
- d) B&C

75. The X-linked recessive disorder of vision is

Options

- a) Tritanopia
- b) Cystic fibrosis
- c) Sickle-cell anaemia
- d) Deuteranopia

76. Assertion (A): Darwinism considers the cumulative effect of fluctuating variations will lead to speciation

Reason (R): Natural selection is a driving force of evolution

Options

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- c) (A) is correct but (R) is not correct
- d) (A) is not correct but (R) is correct

77. In a population of 1000 individuals 28% are homozygous dominant. 48% are heterozygous dominant and remaining are recessive. Find out the dominant frequency

Options

- a) 0.76
- b) 0.52
- c) 0.24
- d) 0.48

78. The desynchronised waves recorded by EEC in persons who are tense are

Options

- a) Delta waves
- b) Theta waves
- c) Alpha waves
- d) Beta waves

79. Match the following

List -1	List - II
A. Ovaprim	I. Membrane protiens
B. Omega-3-fatty acid	II. Pro insulin
C. Cadherins	III. Reduce cholestrol
D. C-Peptide	IV. Synthetic analogue gonadotropin
	V. Exotoxin

The correct answer is
options

A B C D

- a) IV V I II
- b) V III I II
- c) IV III I II
- d) II I III IV

80. Study the following statements

- A. The milk of transgenic cow Rosie contains α -1 antitrypsin
- B. Cancer patients are given alpha interferons to destroy tumors
- C. Attenuated whole agent vaccines act against typhoid
- D. If the apoptosis pathway is damaged the cell dies

Choose the correct statements from the above

Options :

- a) A&C
- b) B & D
- c) B&C
- d) A & D

Physics

81. If L represents inductance. R represents resistance and Q represents electric charge, then match the following.

List - I
Quantity

List - II
Dimensions

A. $\frac{Q^2 R^2}{L}$	I. $M^1 L^2 T^{-3} A^0$
B. $\frac{Q^2 R^3}{L^2}$	II. $M^0 L^0 T^0 A^1$
C. $\frac{Q R^2}{L}$	III. $M^1 L^2 T^{-2} A^0$

D. $\frac{QR}{L}$	IV. $M^1L^2T^{-3}A^{-1}$
-------------------	--------------------------

The correct answer is

Options :

- | | | | | |
|----|-----|-----|----|----|
| | A | B | C | D |
| a) | IV | III | II | I |
| b) | III | I | IV | II |
| c) | IV | III | I | II |
| d) | III | IV | I | II |

82. Two bodies falling freely under gravity from heights 'h' and 'H' cover displacements of $\frac{5h}{9}$ and respectively during last second of their motion. If acceleration due to gravity is 10 ms^{-2} . then H is

Options :

- a) 45 m
- b) 90 m
- c) 180 m
- d) 360 m

83. Assertion (A): When the range of a projectile is maximum. the maximum height reached is 25% of its maximum range.

Reason (R): The maximum height of a projectile is proportional to the square of the velocity of projection.

Options:

- a) Both (A) and (R) are true and (R) is the correct explanation of (A)

- b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- c) (A) is true but (R) is false
- d) (A) is false but (R) is true

84. A particle moves along a circular path of radius R such that its speed ' v ' depends upon distance ' s ' as per

the relation $v = \alpha\sqrt{s}$ where ' α ' is a positive constant. Acceleration of the particle after traversing a distance ' s ' is

Options:

- a) $\alpha^2 \sqrt{\frac{1}{4} - \frac{s^2}{R^2}}$
- b) $\alpha^2 \sqrt{\frac{1}{4} + \frac{s^2}{R^2}}$
- c) $\alpha \sqrt{\frac{1}{2} + \frac{s^2}{R^2}}$
- d) $\alpha^2 \sqrt{\frac{1}{2} + \frac{s^2}{R^2}}$

85. When an empty plastic box of mass 8 kg is placed deep inside water, the box accelerates up with $\frac{g}{4}$. Mass of the sand that should be kept inside the box so that it may accelerate downward with $\frac{g}{3}$ is

(g - acceleration due to gravity)

Options:

- a) 14 kg
- b) 8 kg

- c) 7kg
- d) 10kg

86. The time taken by a block to slide down a smooth inclined plane of inclination 45° is t_1 and the time taken by the same block to slide down a rough inclined plane of coefficient of kinetic friction $\frac{2}{3}$ is t_2 . If the length and angle of inclination of the rough plane are same as those of the smooth plane, then $\frac{t_1}{t_2}$ is

Options:

- a) $\frac{1}{\sqrt{3}}$
- b) $\frac{1}{\sqrt{2}}$
- c) $\frac{1}{2}$
- d) $\frac{2}{\sqrt{3}}$

87. An object of mass 2 kg moving with velocity 20 ms^{-1} collides a stationary metal body of mass 100 g placed on a rough horizontal plane of coefficient of friction 0.5. After the collision, the first body comes to rest and the second body moves on the rough horizontal plane through 100 m before coming to rest. If the energy lost in the collision is absorbed as heat by the second body, then the raise in temperature of the second body is

(specific heat capacity of the metal = $0.2 \text{ J g}^{-1} \text{ K}^{-1}$, $g = 10 \text{ ms}^{-2}$)

Options :

- a) 35.5°C
- b) 12.5°C
- c) 19.5°C
- d) 17.5°C

88. A cyclist rides up a hill with constant velocity. When he exerts an average force of 20 kg-wt on the 50 cm pedal rod. it completes 6 revolutions in 4 s . The power developed by cyclist is

Options :

- a) 0.924 kW
- b) 3.20 kW
- c) 1032 kW
- d) 723.4 kW

89. A billiard ball of diameter 5 cm at rest is hit by a cue at a height 0.5 cm above its centre, If it rolls with a linear velocity of 10 ms^{-1} , then its angular velocity is

Options :

- a) 250 rad s^{-1}
- b) 400 rad s^{-1}
- c) 200 rad s^{-1}
- d) 500 rad s^{-1}

90. A uniform rod AB of mass ' m ' and length ' $2l$ ' is falling freely without rotation under gravity with AB horizontal. Suddenly the end A is fixed when the speed

of the rod is 'V'. The angular speed with which the rod begins to rotate is

Options

- a) $\frac{V}{l}$
- b) $\frac{8V}{3l}$
- c) $\frac{2V}{3l}$
- d) $\frac{3V}{2l}$

91. A disc of radius R is pivoted at the rim and is set for small oscillations. If a simple pendulum has to have the same time period as that of the disc. length of the simple pendulum should be

Options :

- a) $\frac{5R}{4}$
- b) $\frac{2R}{3}$
- c) $\frac{4}{3R}$
- d) $\frac{4}{3R}$

92. Two particles of masses m_1 and m_2 . initially at rest and at a large distance from each other, move under the action of mutual gravitational force. If R is their

separation at any instant their relative velocity of approach at that instant is _____

Options :

- a) $\sqrt{\frac{G(m_1 + m_2)}{R}}$
- b) $\sqrt{\frac{2G(m_1 + m_2)}{R}}$
- c) $\sqrt{\frac{2G(m_1 - m_2)}{R}}$
- d) $\sqrt{\frac{G(m_1 - m_2)}{2R}}$

93. The rubber cord of a catapult has a cross-sectional area 1 mm^2 and unstretched length 10 cm . If it is stretched to a length 12 cm and a body of mass 5 g is projected from it. then the velocity of projection of the body is (Young's modulus of rubber $= 5 \times 10^8 \text{ Nm}^{-2}$)

Options :

- a) 5 ms^{-1}
- b) 10 ms^{-1}
- c) 15 ms^{-1}
- d) 20 ms^{-1}

94. An ornament made by mixing copper and gold weighing 50 g in air. weighs only 46 g in water. If the specific gravities of gold and copper are 20 and 10 respectively, then the mass of copper mixed is

Options :

- a) 30 g
- b) 20 g

- c) 25 g
- d) 15 g

95. Two metal spheres S_1 and S_2 are made of the same material and finish. The mass of S_1 is thrice that of S_2 . Both the spheres are insulated from each other and are heated to the same high temperature and placed in the same room having lower temperature. The ratio of initial rates of cooling of S_1 and S_2 is

Options:

- a) $\left(\frac{1}{3}\right)^{\frac{1}{3}}$
- b) $\frac{1}{\sqrt{3}}$
- c) $\frac{\sqrt{3}}{1}$
- d) $\frac{1}{3}$

96. Ice at 0°C is added to 200 g of water initially at 70°C in a flask, when 50 g of ice has been added to it and is melted completely then the temperature of flask and contents becomes 40°C . When a further 80 g of ice is added and melted completely the temperature of whole system becomes 10°C . Neglecting the heat lost to the surroundings the latent heat of fusion of ice is_____

Options :

- a) 80 cal g^{-1}
- b) 540 cal g^{-1}
- c) 60 cal g^{-1}

d) 90 cal g^{-1}

97. When some amount of heat energy is given to a monatomic gas, the work done by the gas is 70 J. If the same amount of heat energy is given to a diatomic gas, the increase in internal energy of the diatomic gas is _____

Options :

- a) 125 J
- b) 150 J
- c) 175 J
- d) 70 J

98. An ideal gas is confined to an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^x where V is the volume of gas. The value of x is (γ = Ratio of specific heat capacities of the gas)

Options :

- a) $\frac{\gamma + 1}{2}$
- b) $\frac{\gamma - 1}{2}$
- c) $\frac{1 + 2\gamma}{3}$
- d) $\frac{1 + 3\gamma}{5}$

99. For a perfect gas the ratio of specific heat capacities is given by 1.5. The number of degrees of freedom of the molecule is

Options:

- a) 3
- b) 4
- c) 8
- d) 0.5

100. A transverse wave is propagated from the lower end of a rope which is suspended vertically from a rigid support. The nature of the graph drawn between the speed of the wave taken along Y-axis and height from the lower end taken along X-axis is

Options:

- a) Parabola
- b) Straight line parallel to X-axis
- c) Straight line parallel to Y-axis
- d) A sine curve

101. A sufficiently long closed organ pipe has a small hole at its bottom. Initially the pipe is empty. Then if water is poured into the pipe at a constant rate, the fundamental frequency of the air column in the pipe is

Options :

- a) Continuously increases
- b) First increases and then becomes constant
- c) Continuously decreases
- d) First decreases and then becomes constant

102. A thin equiconvex lens made of a material of refractive index $\frac{3}{2}$ has focal length 0.2 m.

When it is immersed in a liquid, it acts like a concave lens of focal length 0.5 m. Then, the refractive index of the liquid is

Options :

- a) $\frac{17}{8}$
- b) $\frac{15}{8}$
- c) $\frac{13}{8}$
- d) $\frac{9}{8}$

103. Two coherent sources of equal intensities produce maximum intensity of 100 units at a point. If the intensity of one of the sources is reduced by 50% by reducing its width, then the intensity of light at the same point will be ____ units.

Options :

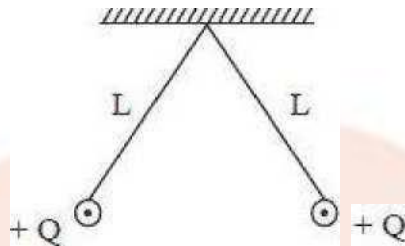
- a) 90.35
- b) 89.86
- c) 67.35
- d) 72.85

104. A system has two charges $2.5 \times 10^{-7} \text{C}$ and $-2.5 \times 10^{-7} \text{C}$ located at points A(0, 0, -15) cm and B(0, 0, 15) cm respectively. Then the total charge and the dipole moment of the system respectively are _____ (in C. Cm)

Options :

- a) 0, 7.5×10^{-8}
- b) 7.5×10^{-8} , 5×10^{-7}
- c) 0, 75×10^{-8}
- d) 5×10^{-7} , 7.5×10^{-8}

105. Two small spheres each having a charge $+Q$ are suspended by two insulating strings of length L from a rigid support as shown in the figure. This setup is taken into a satellite where there is no gravity. Then the tension in each string is



Options:

- a) $\frac{Q^2}{16\pi\epsilon_0 L^2}$
- b) $\frac{Q^2}{8\pi\epsilon_0 L^2}$
- c) $\frac{Q^2}{4\pi\epsilon_0 L^2}$
- d) $\frac{Q^2}{2\pi\epsilon_0 L^2}$

106. Two neutral metal spheres, each of radius 5 cm are placed with a separation of 1 m between their centres. Some charge is transferred from one sphere to the other such that a potential difference of 33.4V is developed between surface of the two spheres. A proton released from rest from the surface of the positively charged sphere strikes the surface of the other sphere with a speed of _____ms⁻¹.

Options:

- a) 2×10^4

- b) 4×10^4
- c) 8×10^4
- d) 16×10^4

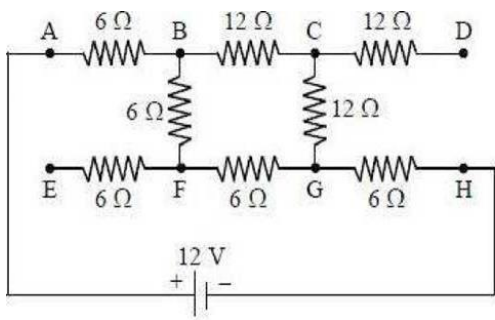
107. A parallel plate air capacitance 'C'. When it is half filled with a dielectric constant 5 as shown in the figure, the percentage increase in the capacitance will be _____



Options:

- a) 86.7
- b) 33.3
- c) 66.7
- d) 200

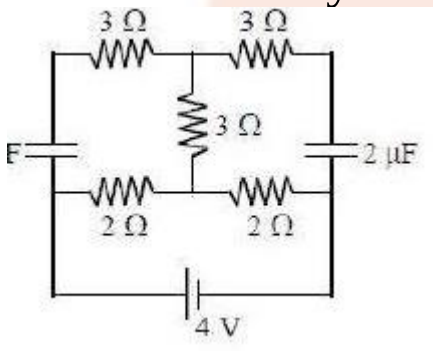
108. Eight resistors are connected as shown in the figure to a battery of emf 12 V. The current through 12Ω resistor between the junctions B and C is _____



Options:

- a) 50mA
- b) 100mA
- c) 150mA
- d) 200mA

109. For the circuit shown in the figure, the power dissipated in the steady state is _____



Options:

- a) 4W
- b) $\frac{1}{4}$ W
- c) 1w
- d) 0.5W

110. When a superconducting loop with zero resistance is moving towards a magnet, the flux linked with the loop

Options:

- a) Increases
- b) Decreases
- c) Remains constant
- d) First increases then decreases

111. A copper wire of diameter 3.2 mm carries a current. If the maximum magnetic field due to this wire is 2.5 mT, then the current in the wire is

Options:

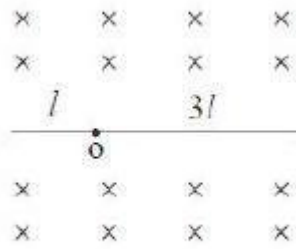
- a) 10A
- b) 40A
- c) 30A
- d) 20A

112. A telephone cable at a place has 5 long straight horizontal wires each carrying a current of 1.4 A from east to west. The angle of dip at the place is 60° and magnetic declination is zero. The resultant magnetic field at a point 3.5 cm below the cable is _____ T.

Options :

- a) 2×10^{-5}
- b) 3×10^{-5}
- c) 4×10^{-5}
- d) 5×10^{-5}

113. A rod of length $4l$ is rotating about an axis passing through it at a point 'O' as shown in the figure with an angular velocity ω in a uniform magnetic field of intensity B which is perpendicular to the plane of rotation. Induced emf between its ends is



Options :

- a) 0
- b) $2Bl^2\omega$
- c) $4Bl^2\omega$
- d) $\frac{Bl^2\omega}{2}$

114..An ac voltage of $100 \text{ V} \cdot \frac{500}{\pi} \text{ Hz}$ is connected across a 20Ω resistor and 15 mH inductor in series. Then the impedance of the circuit is

Options :

- a) 25Ω
- b) 40Ω
- c) 5Ω
- d) 35Ω

115.A laser pointer of power 3 mW creates a spot of 2 mm diameter on a screen. If the screen reflects 57% of the light that strikes it. the radiation pressure on the screen is _____

(speed of light in vacuum = $3 \times 10^8 \text{ ms}^{-1}$)

Options :

- a) $3 \times 10^{-6} \text{Nm}^{-2}$
- b) $4 \times 10^{-6} \text{Nm}^{-2}$
- c) $5 \times 10^{-6} \text{Nm}^{-2}$
- d) $6 \times 10^{-6} \text{Nm}^{-2}$

116..A photon of energy E emits a photoelectron from a metal surface whose work function is W_0 . If this electron enters a uniform magnetic field B in a direction perpendicular to the field and describes a circular path of radius ' r ' then ' r ' is

Options:

- a) $\frac{\sqrt{2m(E - W_0)}}{3eB}$
- b) $\frac{\sqrt{2m(E - W_0)}eB}{3}$
- c) $\frac{\sqrt{2e(E - W_0)}}{mB}$
- d) $\frac{\sqrt{2m(E - W_0)}}{eB}$

117.The radius of the first orbit of hydrogen is r_H . Considering μ^- panicle with mass $207 m_e$ revolving round a proton like m Hydrogen atom, the energy and die radius of the orbit of the μ^- particle when it is in the first orbit are respectively (Assuming proton to be stationary m_e . - rest mass of electron)

Options :

- a) $-13.6 \times 207 eV, \frac{r_H}{207}$
- b) $-207 \times 13.6 eV, 207 r_H$

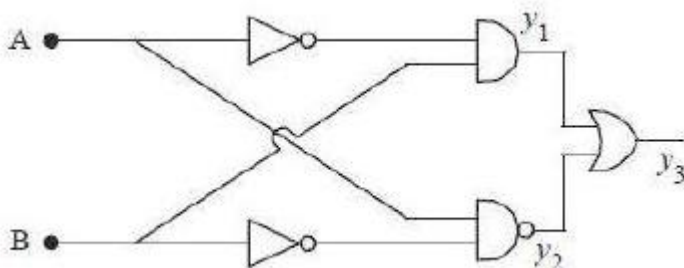
- c) $-\frac{13.6}{207}eV, \frac{r_H}{207}$
d) $-\frac{13.6}{207}eV, 207r_H$

118. A nuclear reactor of power 200 MW uses U^{235} as fuel. If 0.1% of mass of U^{235} is converted into energy. the mass of the fuel consumed per day is

Options :

- a) 192g
b) 19.2 g
c) 0.192 g
d) 1920 g

119. Five logic gates are connected as shown in the figure. If $A = 1$ and $B = 1$ the values of y_1, y_2 and y_3 respectively are



Options :

- a) 0, 1, 1
b) 1, 0, 0
c) 1, 0, 1
d) 1, 1, 0

120. The modulating wave is given by $V_m = 4\sin\omega_m t$ and the carrier wave is given by $V_c = 16 \sin \omega_c t$. The percentage of modulation is

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121. If the wavelength associated with particle A is $5 \times 10^{-8} \text{m}$, then the wavelength associated with particle B having momentum half that of A in 'm' is

Options :

- a) 5×10^{-8}
- b) 1×10^{-7}
- c) 2×10^{-9}
- d) 6×10^{-3}

122. Assertion (A): The kinetic energy of the photo electrons increases with the increase of frequency of incident light

Reason (R): If frequency of incident light is less than the threshold frequency electrons

are not emitted from metal surface

The correct answer is

Options :

- a) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- b) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
- c) (A) correct but (R) is not correct
- d) (A) is not correct but (R) is correct

123. The change in properties which are common to elements of both groups 1 and 17 in the periodic table are

- a) Ionization enthalpy decreases down the groups
- b) Reactivity decreases down the groups

- c) Atomic radius increases down the groups
 d) Electronegativity decreases down the groups

Options :

- a) a,c,d
 b) a,b,c,d
 c) a,b,c
 d) b,c,d

124. Which one of the following statements is not correct?

Options:

- a) s-orbital does not form a π -bond
 b) Energy of resonance hybrid of the O_3 is lower than that of either of the two canonical forms
 c) o-nitrophenol is involved in intermolecular hydrogen bond
 d) Dipole moment of NF_3 is higher than NF_3

125. The correct order of type of hybridisation and shape of the following species is

- a) $[Ni(CN)_4]^{2-}$
 b) CF_4
 c) BrF_5
 d) $[CrF_6]^{3-}$

Options :

a	b	c	d
a) sp^3 Tetrahedral	sp^3d Trigonal bipyramidal	sp^3d^2 Octahedral	sp^3d^2 Square pyramidal
b) dsp^2	sp^3d	sp^3d^2	d^2sp^3

Square planar	Square pyramidal	Trigonal bi pyramidal	Octahedral
c) dsp^2 Square planar	sp^3d T-shape	sp^3d^2 Square pyramidal	sp^3d^2 Octahedral
d) sp^3 Tetrahedral	sp^3d T-shape	sp^3d^2 Square pyramidal	sp^3d^2 Octahedral

126. 3.2g of oxygen and 0.2 g of hydrogen are placed in a 1.12 L flask at 0°C . The total pressure of the gaseous mixture in atm is

(assume oxygen and hydrogen as ideal gases)

Options :

- a) 1
- b) 4
- c) 3
- d) 2

127. When 50g of sulphur was burnt in a 4*5 b of sulphur was left over. The volume of air in litres required at STP containing 21% oxygen by volume is

Options :

- a) 33.6
- b) 11.2
- c) 160
- d) 22.4

128. Match the following

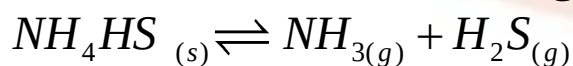
List -I			List -II
ΔH	ΔS	ΔG	(Description of reaction)
A) -	+	-	I) Spontaneous at low T
B) -	-	-	II) Non spontaneous at high T
C) -	-	+	III) Spontaneous at all T
D) +	+	-	IV) Spontaneous at high T
			V) Non spontaneous at all temperatures

The correct answer is

Options:

- | | | | | |
|----|-----|----|-----|-----|
| | A | B | C | D |
| a) | III | II | I | V |
| b) | V | II | I | III |
| c) | IV | I | III | II |
| d) | III | I | II | IV |

129. Observe the following reaction



If the total pressure for this reaction at equilibrium is 30 atm. the value of K_p (in atm^2) is

Options :

- a) 225
- b) 15
- c) 30

d) 125

130. If pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4 respectively, the pH of their salt solution (BA) is

Options:

- a) 7.2
- b) 7.0
- c) 6.9
- d) 7.6

131. What volume of oxygen in 'L' at STP is liberated when acidified $KMnO_4$ solution containing 0.1 mol of $KMnO_4$ is added to excess H_2O_2 solution?

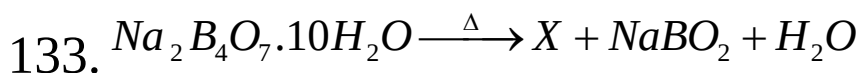
Options :

- a) 5.6
- b) 11.2
- c) 22.4
- d) 2.24

132. On dissolving moderate amount of sodium metal in liquid ammonia at low temperature, which one of the following does not occur ?

Options :

- a) Blue colored solution is obtained
- b) This solution has paramagnetic property
- c) This solution becomes good conductor of electricity
- d) This solution remains diamagnetic





X and Y are

Options :

- a) $\text{Na}_3\text{BO}_3, \text{Co}(\text{BO}_2)_2$
- b) $\text{Na}_2\text{B}_4\text{O}_7, \text{Co}(\text{BO}_2)_2$
- c) $\text{B}_2\text{O}_3, \text{Co}(\text{BO}_2)_2$
- d) $\text{B}_2\text{O}_3, \text{Co}(\text{BO}_2)_2$

134. Which of the following statement is not correct with respect to group 14 elements?

Options :

- a) The stability of +4 oxidation state decreases in the group
- b) The common oxidation state for carbon and silicon is +4
- c) Pb^{4+} is more stable than Pb^{2+} ion
- d) The stability of +2 oxidation state increases in the group due to inert pair effect

135. Which of the following statement is not correct?

Options :

- a) Lower the concentration of dissolved oxygen, the more polluted is the water sample
- b) The tolerable limit of lead in drinking water is 50 ppb
- c) Water is considered pure if it has BOD less than 5 ppm
- d) In COD determination, the pollutants resistant to microbial oxidation are not oxidised by oxidising agent like $\text{K}_2\text{Cr}_2\text{O}_7$

136. 0.5g of an organic compound on analysis by Kjeldahl's method gave enough ammonia to just neutralise 20mL of

0.5M H₂SO₄. The percentage of nitrogen in the compound is

Options :

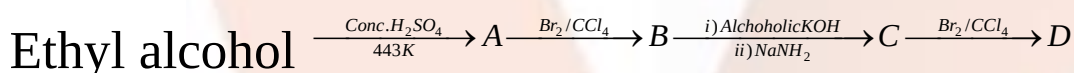
- a) 56
- b) 28
- c) 42
- d) 14

137. Which of the following is oxidized by KMnO₄ solution?

Options :

- a) CH₃CH₂CH₃
- b) CH₄
- c) CH₃CH₃
- d) (CH₃)₃CH₂

138. What is D in this reaction sequence?



Options :

- a) CH $\equiv$ CH
- b) CHBr₂-CHBr₂
- c) CH₂Br-CH₂Br
- d) CH₃-CHBr₂

139. In a cubic close packed structure of mixed oxide, the lattice is made of oxide ions. One eighth of tetrahedral voids are occupied by cation A while one half of octahedral voids are occupied by cation B. The formula of the oxide is

Options:

- a) ABO
- b) ABO_3
- c) $\text{A}_7\text{B}_2\text{O}_7$
- d) AB_2O_4

140. Which one of the following is correct for solutions of components of A and B to follow Raoult's law?

Options :

- a) A-B attractive force is greater than A-A and B-B attractive forces
- b) A-B attractive force is less than A-A and B-B attractive forces
- c) A-B attractive force remains same as A-A and B-B attractive forces
- d) Volume of solution is different from sum of volumes of A and B components

141. How many grams of sucrose should be dissolved in 100g water in order to produce a solution having difference between the freezing point and the boiling point temperatures as 105.0°C ?

Options :

- a) 4.2
- b) 72
- c) 342
- d) 460

142. At $T(\text{K})$ the molar ionic conductivities of NH_4^+ and OH^- at infinite dilution are 72 and $198 \text{ S cm}^2 \text{ mol}^{-1}$ respectively.

The molar conductivity of 0.01M NH_4OH solution at the same temperature is found to be $9 \text{ S cm}^2 \text{ mol}^{-1}$. The percentage dissociation of NH_4OH at this concentration is

Options :

- a) 3.33
- b) 7.14
- c) 12.5
- d) 4.54

143. When the concentration of a reactant in the reaction. $A \rightarrow B$ is increased by 8 times, the rate increases by 2 times. The order of reaction is

Options :

- a) 2
- b) 3
- c) $1/3$
- d) 4

144. If the gold number of potato starch is 25. how many grams of it is required to prevent coagulation of 100mL of gold sol on adding 10mL of 10% NaCl solution'?

Options :

- a) 0.30
- b) 0.50
- c) 0.25
- d) 0.40

145. Which one of the following methods is used for refining metals of low boiling point when they contain impurities of high boiling point?

Options :

- a) Zone refining
- b) Poling
- c) Distillation
- d) Froth floatation

146. The chemicals used in Holme's signals are

Options :

- a) Calcium nitride and calcium carbide
- b) Magnesium nitride and calcium phosphide
- c) Calcium carbide and calcium phosphide
- d) Calcium chloride and calcium carbonate

147. Consider the following reactions

$X(aq) + H_2SO_4(dil) \rightarrow Y$. colorless gas with pungent smell

$Y + MnO_4^- + H_2O \rightarrow Z$. colorless solution

Identify X and Y in the above reactions

Options :

- a) SO_3^{2-} , SO_2
- b) NO_3^- , NO
- c) S^{2-} , H_2S
- d) Cl^- , HCl

148. Which one of the following reactions of Xenon compounds is not feasible?

Options :

- a) $XeF_4 + SbF_5 \rightarrow [SbF_4]^+ [XeF_5]^-$
- b) $3XeF_4 + 6H_2O \rightarrow 2Xe + XeO_3 + 12HF + \frac{3}{2} O_2$

- c) $2\text{XeF}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
d) $\text{XeF}_6 + \text{RbF} \rightarrow \text{Rb}[\text{XeF}_7]$

149. The number of unpaired electrons in $[\text{Pr}(\text{CN})_4]^{2-}$ in having square planar geometry (At. No. of Pt = 78) is

Options :

- a) 0
b) 1
c) 2
d) 3

150. Which one of the following is the correct order of ionic radii?

Options :

- a) $\text{Ce}^{+3} < \text{La}^{+3} < \text{Pm}^{+3} < \text{Yb}^{+3}$
b) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3}$
c) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$
d) $\text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3} < \text{Yb}^{+3}$

151. Which of the following are biodegradable polymers?

- a) PVC
b) PHBV
c) Bakelite
d) Nylon 2 - Nylon 6

Options :

- a) a,b
b) b,c
c) c,d
d) b,d

152. Which of the following does not exist as a zwitter ion?

Options:

- a) Glycine
- b) Alanine
- c) Sulphanilic acid
- d) D-glucose

153. Which detergent is used as a germicide?

Options:

- a) Sodium lauryl sulphate
- b) Cetyltrimethyl ammonium bromide
- c) Glyceryl palmitate
- d) Sodium dodecyl benzene sulphonate

154. Consider the following reaction

Br



Which of the following statements are correct?

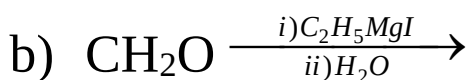
- a) I is the major product of the reaction
- b) II is the major product of the reaction
- c) Formation of I is in accordance with Saytzeff rule
- d) II is more stable because it is more substituted

Options :

- a) a,c
- b) b,c
- c) a,b

d) b,d

155. Which one of the following reactions will yield 2-propanal?



Options :

a) a,b

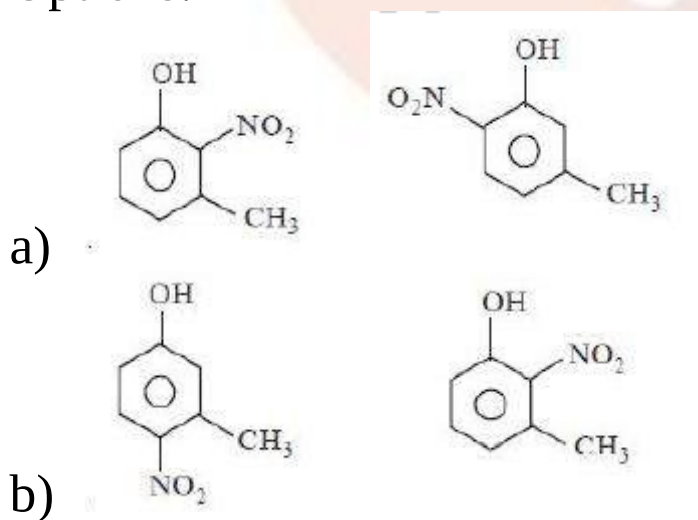
b) b,c

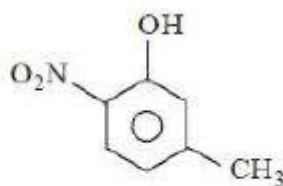
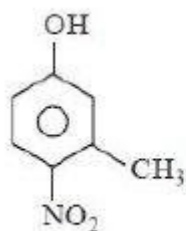
c) a,c

d) c,d

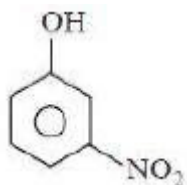
156. The major product/s expected from the mononitration of 3-methyl phenol is

Options:





c)



d)

157. A mixture of benzaldehyde and acetophenone on heating with dilute NaOH solution gives

Options :

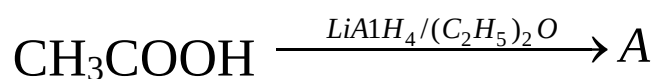
- a) Benzophenone
- b) Benzalacetophenone
- c) Phenyl benzoate
- d) Benzyl alcohol and sodium benzoate

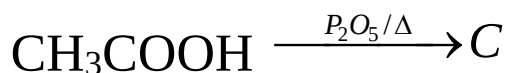
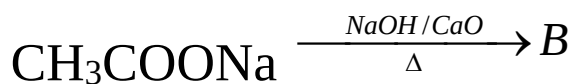
158. Which of the following compounds give a yellow precipitate with iodine and aqueous NaOH?

Options:

- a) a,b
- b) b,c
- c) c,d
- d) a,c

159. What are A, B, C and D in the following reactions?





A

B

C

D

Options:

a) CH ₃ CHO	CH ₃ CH ₃	(CH ₃ CO) ₂ O	CH ₃ COONH ₄
b) CH ₃ CH ₂ OH	CH ₄	(CH ₃ CO) ₂ O	CH ₃ CONH ₂
c) CH ₃ CH ₂ OH	CH ₃ CH ₃	(CH ₃ CO) ₂ O	CH ₃ CONH ₂
d) CH ₃ CHO	CH ₃ COCH ₃	CH ₃ OCH ₃	CH ₃ COONH ₄

160. Which of the following statements are correct?

- The basic strength of ethyl substituted amines in aqueous solution is, (C₂H₅)₃N > (C₂H₅)₂NH > (C₂H₅)₂NH₂
- Aniline has five resonance structures but anilinium ion has only two resonance structures
- Presence of -OCH₃ group on ring increases the basic strength of aniline
- Presence of -NO₂ group on ring decreases the basic strength of aniline

Options:

- a, b, c
- b, c, d
- a, c, d
- a, b, d